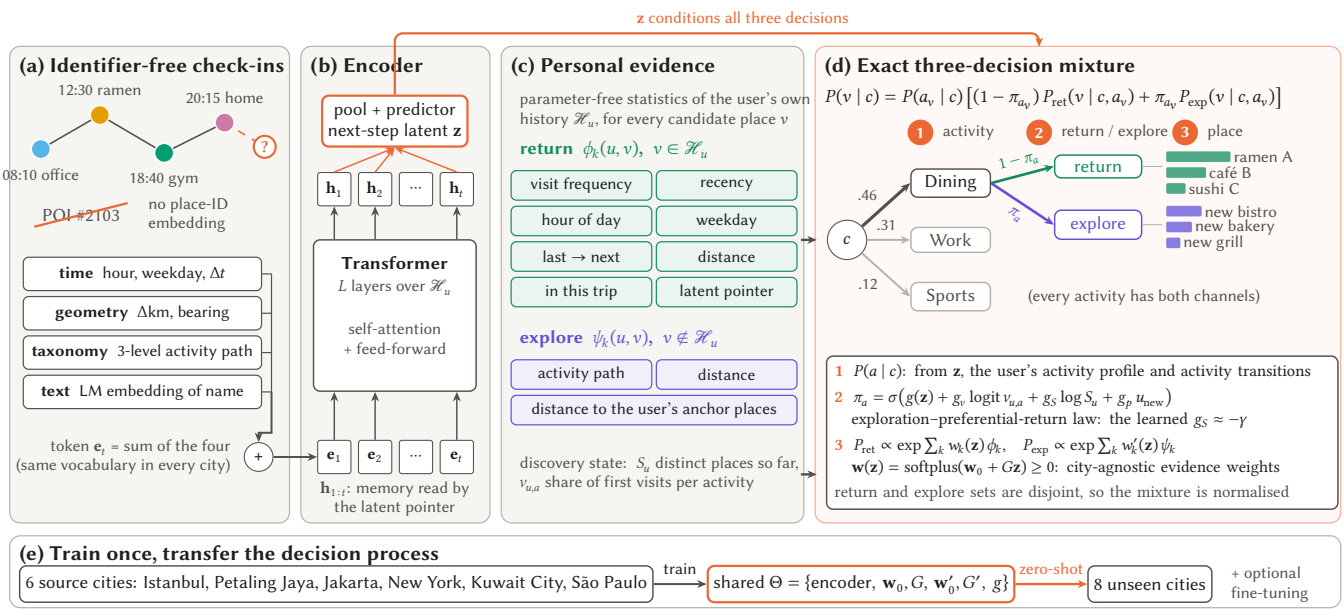
z conditions all three decisions
(a) Identifier-free check-ins
12:30 ramen
20:15 home
08:10 office
18:40 gym
?
POI #2103
no place-ID embedding
time hour, weekday, Δt
geometry Δkm, bearing
taxonomy 3-level activity path
text LM embedding of name
token e_t = sum of the four (same vocabulary in every city)
(b) Encoder
pool + predictor next-step latent z
h_1 h_2 ··· h_t
Transformer
L layers over H_u
self-attention + feed-forward
e_1 e_2 ··· e_t
h_{1:i}: memory read by the latent pointer
(c) Personal evidence
parameter-free statistics of the user's own history H_u, for every candidate place v
return φ_k(u, v), v ∈ H_u
visit frequency
recency
hour of day
weekday
last → next
distance
in this trip
latent pointer
explore ψ_k(u, v), v ∉ H_u
activity path
distance
distance to the user's anchor places
discovery state: S_u distinct places so far, ν_{u,a} share of first visits per activity
(d) Exact three-decision mixture
P(v | c) = P(a_v | c) [(1 − π_{a_v}) P_ret(v | c, a_v) + π_{a_v} P_exp(v | c, a_v)]
1 activity
2 return / explore
3 place
c
.46
.31
.12
Dining
Work
Sports
1 − π_a
π_a
return
explore
ramen A
café B
sushi C
new bistro
new bakery
new grill
(every activity has both channels)
1 P(a | c): from z, the user's activity profile and activity transitions
2 π_a = σ(g(z) + g_v logit ν_{u,a} + g_S log S_u + g_p u_new)
exploration–preferential-return law: the learned g_S ≈ −γ
3 P_ret ∝ exp Σ_k w_k(z) φ_k, P_exp ∝ exp Σ_k w'_k(z) ψ_k
w(z) = softplus(w_0 + Gz) ≥ 0: city-agnostic evidence weights
return and explore sets are disjoint, so the mixture is normalised
(e) Train once, transfer the decision process
6 source cities: Istanbul, Petaling Jaya, Jakarta, New York, Kuwait City, São Paulo
train
shared Θ = {encoder, w_0, G, w'_0, G', g}
zero-shot
8 unseen cities
+ optional fine-tuning